

	Bounded	UnBounded
Can do it	$dp[i-1][j]$ OR $dp[i-1][j - \text{array}[i-1]]$	$dp[i-1][j]$ OR $dp[i][j - \text{array}[i-1]]$
CountWays	$dp[i-1][j] + dp[i-1][j - \text{array}[i-1]]$	$dp[i-1][j] + dp[i][j - \text{array}[i-1]]$
Min Max	$\min(dp[i-1][j], 1 + dp[i-1][j - \text{array}[i-1]])$	$\min(dp[i-1][j], 1 + dp[i][j - \text{array}[i-1]])$

DP Type	Count Ways	Min / Max	Possible / Not	Probability
Sequential DP	Fibonacci Ways, Climbing Stairs	Min Cost Climbing Stairs, House Painting	Valid Step Sequence, Sequential Constraint Validation	Sequential Survival Probability, Reliability Chain
Decision DP	Target Sum Ways, Count Subsequences	House Robber, Delete and Earn	Subset Sum, Partition Equal Subset Sum	Pick/Skip Success Probability, Optimal Decision Probability
Contiguous DP	Count Alternating Subarrays, Count Increasing Segments	Maximum Subarray Sum, Maximum Product Subarray	Continuous Segment Validation, Consecutive Valid Chain	Consecutive Event Probability, Continuous Signal Detection
Prefix DP	Decode Ways, Count Prefix Decompositions	Maximum Prefix Score, Best Prefix Gain	Word Break, Prefix Dictionary Match	Prefix Survival Probability, Running Success Probability
Jump DP	Ways to Reach End, Dice Throw Ways	Minimum Jumps, Frog Jump Min Energy	Jump Game, Reach End Possible	Random Walk Probability, Frog Position Probability
State DP	Attendance Record Count, Stock Trading Count	Stock Buy Sell, Vacation Planning	Reachability with Modes, Task Scheduling States	Weather Transition Probability, Markov State Probability
Partition DP	Integer Break Ways, Count Rod Cutting Combinations	Rod Cutting, Minimum Cost Partition	Word Break, Palindrome Partition Possible	Segment Reliability Probability, Partition Event Probability

DP Type	Bounded Transition	Unbounded Transition
Sequential DP	$dp[i] = f(dp[i-1])$	Usually same as bounded
Decision DP	$dp[i] = \text{best}(\text{skip}, \text{take} \setminus \text{once})$	$dp[i] = \text{best}(\text{skip}, \text{reuse} \setminus \text{take})$
Contiguous DP	Usually bounded naturally	Rarely unbounded
Prefix DP	Prefix used once	Prefix patterns reusable
Jump DP	$dp[i] = \text{combine}(dp[j])$	$dp[i] = \text{combine}(dp[i-\text{step}])$
State DP	Transition to new state once	State revisitable
Partition DP	Partition choices used once	Partition reusable repeatedly

DP Type	Count Ways	Min / Max	Possible / Not	Probability
Sequential DP	$dp[i] = dp[i-1] + \text{ways}(i)$	$dp[i] = \text{best}(dp[i-1] + \text{value}(i))$	$dp[i] = dp[i-1] \wedge \text{valid}(i)$	$dp[i] = dp[i-1] \times P(i)$
Decision DP	$dp[i] = \text{take} + \text{skip}$	$dp[i] = \text{best}(\text{take}, \text{skip})$	$dp[i] = \text{take} \vee \text{skip}$	$dp[i] = P(\text{take}) + P(\text{skip})$
Contiguous DP	$dp[i] = dp[i-1] + \text{newWays}$	$dp[i] = \text{best}(\text{arr}[i], \text{arr}[i] + dp[i-1])$	$dp[i] = \text{extend} \vee \text{restart}$	$dp[i] = \text{combine}(dp[i-1], P(i))$
Prefix DP	$dp[i] = dp[i-1] + \text{prefixWays}$	$dp[i] = \text{best}(dp[i-1], \text{prefix}(i))$	$dp[i] = \text{possible}(\text{prefix}(i))$	$dp[i] = \text{combine}(dp[i-1], \text{prefixProb}(i))$
Jump DP	$dp[i] = \sum dp[j]$	$dp[i] = \text{best}(dp[j] + \text{cost})$	$dp[i] = \vee dp[j]$	$dp[i] = \text{combine}(P(j))$
State DP	$dp[i][s] = \sum dp[\text{prevState}]$	$dp[i][s] = \text{best}(dp[\text{prevState}] + \text{value})$	$dp[i][s] = \text{reachable}(\text{prevState})$	$dp[i][s] = \text{transitionProb}(\text{prevState})$
Partition DP	$dp[i] = \sum dp[j]$	$dp[i] = \text{best}(dp[j] + \text{cost}(j+1, i))$	$dp[i] = \text{possible}(dp[j])$	$dp[i] = \text{combine}(P(\text{partitions}))$